

Requirements Audit — Multi-Branch Retail & Inventory Management System

This document maps **every requirement** from the project specification against what currently exists in the codebase

—  = fulfilled,  = partially fulfilled,  = missing/not found.

Phase 1: Setup & Database Design

#	Requirement	Status	Evidence
1	Git repo with .gitignore and README		.gitignore covers Python, Node.js, env files, OS files. README.md present with features, setup, and tech stack.
2	Virtual environment and dependencies		requirements.txt with Flask, SQLAlchemy, Flask-JWT-Extended, Flask-CORS, psycpg.
3	Database schema — all 9 required tables		users, branches, categories, products, inventory, sales_transactions, transaction_items, stock_transfers, inventory_adjustments — all present as SQLAlchemy models. Plus bonus suppliers table.
4	SQLAlchemy ORM models		Ten model files in backend/app/models/.
5	Database initialization script with sample data		seed.py created and tested.

Schema Detail Check

Table	Required Columns	Status	Notes
users	user_id, name, email, password_hash, role, phone, branch_id, created_at		Has all required + extras (employee_id, interview_status, score, banking info). Name split into first_name/last_name (acceptable variant).
branches	branch_id, name, address, city, state, postal_code, phone, manager_id, status, created_at		All present + branch_code, upi_id.
categories	category_id, name, description, parent_category_id		Exact match.
products	product_id, name, sku, barcode, category_id, description, unit_price, cost_price, image_path, created_at		All present + extras (discount_percent, gst_percent, mfg_date, expiry_date, is_big1, supplier_id, size, unit).
inventory	inventory_id, product_id, branch_id, quantity,		Exact match.

	min_threshold, max_threshold, last_updated		
sales_transactions	transaction_id, branch_id, staff_id, total_amount, payment_method, transaction_date, status	✓	All present + transaction_uuid, invoice_number, discount.
transaction_items	item_id, transaction_id, product_id, quantity, unit_price, subtotal	✓	Exact match.
stock_transfers	transfer_id, product_id, from_branch_id, to_branch_id, quantity, request_date, status, approved_by, completed_date, notes	✓	Exact match.
inventory_adjustments	adjustment_id, product_id, branch_id, adjustment_type, quantity, reason, adjusted_by, adjustment_date	✓	All present + supplier_id.

Phase 2: Backend API Development

Authentication APIs

Endpoint	Required	Status	Implementation
POST /api/auth/register	✓	✓	auth.py → register()
POST /api/auth/login	✓	✓	auth.py → login() with JWT token generation
GET /api/auth/profile	✓	✓	auth.py → profile()
PUT /api/auth/profile	✓	✓	auth.py → update_profile()
Password recovery	✓	✓	forgot_password() + reset_password() via OTP email — exceeds "simulated" requirement.

Branch APIs

Endpoint	Status	Implementation
GET /api/branches	✓	branches.py → get_branches()
GET /api/branches/<id>	✓	branches.py → get_branch()
POST /api/branches (Admin)	✓	branches.py → create_branch() with @roles_required("admin")
PUT /api/branches/<id>	✓	branches.py → update_branch()
GET /api/branches/<id>/inventory	✓	branches.py → get_branch_inventory()

Branch performance metrics in GET		Basic branch details returned, but performance metrics are in the admin dashboard stats instead of the individual branch endpoint.
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Product APIs

Endpoint	Status	Implementation
GET /api/products (pagination)		product.py → get_products() with filtering
GET /api/products/<id>		product.py → get_product()
POST /api/products (Manager/Admin)		product.py → create_product()
PUT /api/products/<id>		product.py → update_product()
DELETE /api/products/<id>		product.py → delete_product()
POST /api/products/<id>/image		Image upload handled inline within create/update , not as a separate endpoint. Functionally equivalent.
Bonus: Bulk delete, Excel import, GST bulk update, sample template download		Extra features beyond requirements.

Inventory APIs

Endpoint	Status	Implementation
GET /api/inventory/branch/<branch_id>		inventory.py → get_inventory_by_branch()
PUT /api/inventory/<id>		inventory.py → update_inventory()
POST /api/inventory/adjust		inventory.py → adjust_inventory()
GET /api/inventory/low-stock		inventory.py → get_low_stock()
GET /api/inventory/product/<product_id>		Not as a dedicated endpoint. Product inventory across branches is handled in admin inventory reports.

Sales/POS APIs

Endpoint	Status	Implementation
POST /api/sales		sales.py → create_sale()
GET /api/sales/branch/<branch_id>		sales.py → get_sales_by_branch()
GET /api/sales/<id>		sales.py → get_sale()
POST /api/sales/<id>/refund		sales.py → refund_sale()
GET /api/sales/daily-summary		sales.py → get_daily_summary()

Void transaction	✓	sales.py → void_sale()
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Stock Transfer APIs

Endpoint	Status	Implementation
GET /api/transfers	✓	transfer.py → get_transfers()
POST /api/transfers	✓	transfer.py → create_transfer()
PUT /api/transfers/<id>/approve	✓	transfer.py → approve_transfer()
PUT /api/transfers/<id>/complete	✓	transfer.py → complete_transfer()
GET /api/transfers/<id>	✓	transfer.py → get_transfer()
Bonus: Reject + Cancel	✓	reject_transfer() + cancel_transfer() — extra functionality.

Admin APIs

Endpoint	Status	Implementation
GET /api/admin/users	✓	admin.py → get_users()
PUT /api/admin/users/<id>	✓	admin.py → update_user()
GET /api/admin/stats	✓	admin.py → get_dashboard_stats()
GET /api/admin/reports	✓	admin.py → sales_report(), inventory_report(), top_products_report()
User deactivate/reactivate	✓	delete_user() (soft delete) + reactivate_user()
Create user	✓	admin.py → create_user()
Approve user	✓	admin.py → approve_user()
Bonus: Stock rebalancing suggestions	✓	admin.py → rebalance_suggestions()

Backend Infra Requirements

Requirement	Status	Evidence
Proper error handling with HTTP codes	✓	Routes use 400, 401, 403, 404, 409, 500 codes. try-except in sales, product, branches, auth, admin.
Input validation for API endpoints	✓	Section headers and basic checks in routes.
Role-based route protection	✓	@roles_required() decorator in decorators.py used across routes.
CORS enabled	✓	Flask-CORS in extensions + requirements.

Phase 3: Frontend (HTML/CSS/JS)

The project uses **React (Phase 4)** instead of raw HTML/CSS/JS. This effectively combines Phases 3 and 4.

Requirement	Status	Notes
Responsive layouts	✓	Bootstrap 5 imported + custom CSS with responsive design.
Login page with validation	✓	Login.jsx with client-side validation and error display.
Registration page	⚠	Register.jsx exists but is commented out in App.jsx — disabled for "private enterprise" mode. The component still exists.
Staff POS interface	✓	staffPOS.jsx (44KB) — comprehensive POS with product search, cart, payment methods.
Receipt generation	✓	staffReceipts.jsx — receipt display and print.
Staff inventory check	✓	staffInventory.jsx.
Staff daily sales view	✓	staffDashboardHome.jsx.
Manager dashboard	✓	ManagerDashboard.jsx + ManagerDashboardHome.jsx with overview cards.
Manager inventory status	✓	ManagerInventory.jsx.
Manager transactions	✓	ManagerTransactions.jsx.
Manager transfers	✓	ManagerTransfers.jsx.
Manager reports	✓	ManagerReports.jsx.
Manager staff activity	✓	ManagerStaffActivity.jsx.
Admin multi-branch charts	✓	AdminDashboard.jsx + AdminReports.jsx with Chart.js & ApexCharts.
Admin inventory overview	✓	AdminInventory.jsx (69KB — comprehensive inventory management).
Admin user management	✓	User management in admin dashboard with create/edit/deactivate.
Admin branch management	✓	AdminBranches.jsx.
Admin product catalog	✓	Product management via admin inventory (import, bulk ops, GST).

Admin analytics with charts	✓	Chart.js + ApexCharts in use.
JWT stored in localStorage	✓	authService.js + AuthContext.jsx handle token storage.
Logout functionality	✓	Present in auth flow.
Barcode scanning (simulated)	⚠	Barcode field exists in product model and search, but no barcode scanner simulation UI found (no camera/scan interface). Products can be searched by barcode text.

Phase 4: React Frontend Integration

Requirement	Status	Evidence
React project (CRA or Vite)	✓	Vite project (@vitejs/plugin-react).
App.js with routing	✓	App.jsx with React Router v7.
Login.js, Register.js	✓	Both exist. Register disabled in routing but component present.
StaffDashboard.js	✓	StaffDashboard.jsx.
ManagerDashboard.js	✓	ManagerDashboard.jsx.
AdminDashboard.js	✓	AdminDashboard.jsx.
Navbar.js	✓	Navbar.jsx.
POSInterface.js	✓	staffPOS.jsx (44KB POS component).
ProductCard.js	⚠	No standalone ProductCard.js; product display is inline within inventory/POS components.
InventoryTable.js	⚠	No standalone component; inventory tables are inline in AdminInventory.jsx, ManagerInventory.jsx, staffInventory.jsx.
TransferForm.js	⚠	Transfer forms are inline within ManagerTransfers.jsx and adminStockTransfer.jsx.
SalesChart.js	✓	SalesChart.jsx.
UserTable.js	✓	UserTable.jsx.
BranchSelector.js	⚠	Branch selection exists inline in various components, not as standalone BranchSelector.jsx.
React Router	✓	react-router-dom v7 with protected routes.
React hooks (useState, useEffect)	✓	Used throughout all components.
Axios/Fetch for API calls	✓	axios in services/api.js + services/authService.js.

Protected routes by role	✓	ProtectedRoute.jsx component used in App.jsx.
Props between components	✓	Used throughout.
Reusable components	✓	StatCard, Toast, ConfirmModal, LiveClock, DashboardFAQ, SalesChart, etc.
Loading states & error handling	✓	Present across dashboard and data-fetching components.
Bootstrap styling	✓	Bootstrap 5 imported in main.jsx.
Cart context for POS	✓	CartContext.jsx in context/.

Phase 5: Advanced Features & Testing

Requirement	Status	Evidence
File Upload — Flask multipart/form-data	✓	Product image uploads handled; uploads/ directory exists.
File type/size validation	⚠	Needs verification — basic handling present but explicit size limits unclear.
Product images with fallback	⚠	Images stored by path, but no explicit fallback placeholder logic found.
Search & Filter — product by name/SKU/barcode	✓	Product search in POS and inventory routes.
Filter sales by date/payment/branch	✓	get_sales() with filters; AdminReports.jsx with calendar filtering.
Filter inventory by stock/category	✓	Inventory routes support filtering.
Sort functionality for tables	✓	Present in admin/manager inventory views.
Real-time Features — low-stock alerts	✓	Low-stock endpoint + dashboard alerts.
Stock level updates after sales	✓	create_sale() decrements inventory.
Automatic inventory adjustment on transfers	✓	approve_transfer()/complete_transfer() adjusts inventory.
Daily sales summary	✓	get_daily_summary().
Reporting — Sales report generation	✓	AdminReports.jsx (31KB) + backend report routes.
PDF/Excel export	⚠	Excel import exists (openpyxl in requirements). PDF export references found in AdminDashboard.jsx but no

		ReportLab/WeasyPrint in backend requirements. Export may be frontend-only (print-to-PDF).
Inventory valuation reports	✓	<code>inventory_report()</code> in admin routes.
Profit margin calculations	⚠	<code>cost_price</code> exists in product model, but dedicated margin analysis route is not clearly dedicated.
Top products report	✓	<code>top_products_report()</code> .
Exception Handling — try-except in routes	✓	Present in sales, product, branches, auth, admin routes.
Custom error pages (404, 500)	✓	Global handlers in <code>app/__init__.py</code> .
Python logging module	✓	Configured in <code>app/__init__.py</code> with file logging.
Transaction rollback on failures	✓	<code>db.session.rollback()</code> in several routes.
Security — Password hashing	✓	Werkzeug <code>generate_password_hash</code> / <code>check_password_hash</code> .
Input sanitization (SQL injection prevention)	✓	SQLAlchemy ORM provides parameterized queries.
CSRF protection	⚠	JWT-based API (stateless, no cookies by default), so CSRF is largely N/A. No explicit CSRF tokens for forms.
Secure file upload validation	⚠	Basic handling but no explicit <code>werkzeug.utils.secure_filename</code> verification found.
Testing — API tests (Postman / pytest)	✗	No test files found — no <code>tests/</code> directory, no <code>test_*.py</code> files, no Postman collection.
Test users for each role	✗	No seed script or test fixtures.

Phase 6: Deployment & Documentation

Requirement	Status	Evidence
README with project description	✓	Detailed README with features, roles matrix, tech stack, setup.
Installation instructions	✓	Backend + Frontend setup instructions present.
API documentation in README	✓	Comprehensive section added to README.
Database schema diagram	✓	Summary section added to README.
Screenshots of key pages	✗	No screenshots included.

User roles & permissions matrix	✓	Table in README.
Separate config file for env vars	✓	config.py + .env file.
Modular Flask Blueprints	✓	10 blueprints in routes/__init__.py.
requirements.txt	✓	Present with all dependencies.
Environment-specific settings	⚠	Single config class. No DevelopmentConfig / ProductionConfig separation.
Docstrings in functions	⚠	Module-level docstrings present in route files. Individual function docstrings are missing.
Commented complex logic	✓	Section headers and inline comments throughout.

Summary Scorecard

Category	✓ Done	⚠ Partial	✗ Missing	Total
Phase 1: Database	4	0	1	5
Phase 2: Backend APIs	28	3	0	31
Phase 3 & 4: Frontend	30	6	0	36
Phase 5: Advanced	11	5	3	19
Phase 6: Documentation	6	2	3	11
TOTAL	79	16	7	102

Overall Completion: ~77% fully done, ~16% partial, ~7% missing.

🔴 Items That Need Work (Prioritized)

Must Fix (✗ Missing)

1. **Database seed script** — Create `seed.py` with sample data for demo (users of each role, branches, products, inventory, transactions)
2. **Python logging** — Add `logging` module to backend for error/activity logging to a file
3. **No tests** — Create at minimum a Postman collection or pytest test suite for API endpoints
4. **Custom error pages** — Add 404/500 error handlers in Flask app
5. **API documentation** — Add endpoint docs (URL, method, params, responses) to README
6. **DB schema diagram** — Add an ER diagram to README
7. **Screenshots** — Add screenshots of key pages to README

Should Improve (⚠ Partial)

8. **Register page** — Re-enable or document why it's disabled. The requirement asks for it.
9. **Input validation** — Flesh out `validators.py` with proper validation logic
10. **Profit margin analysis** — Add a dedicated profit margin report endpoint

11. **Product inventory across branches** — Add `GET /api/inventory/product/<product_id>` endpoint
12. **PDF export** — Add server-side PDF generation (e.g., using `reportlab` or `weasyprint`)
13. **Standalone React components** — Extract `ProductCard` , `InventoryTable` , `TransferForm` , `BranchSelector` as separate files (even if functionality exists inline)
14. **Environment-specific config** — Add `DevelopmentConfig` / `ProductionConfig` classes
15. **Function docstrings** — Add docstrings to all route handler functions